

Лабораторная работа 7

Супонина Анастасия Павловна

6 декабря 2025

РУДН, Москва, Россия

Лабораторная работа 7

Цель работы

Изучить способы создания презентации и постеров в LaTeX.

Выполнение лабораторной работы

Презентация в beamer

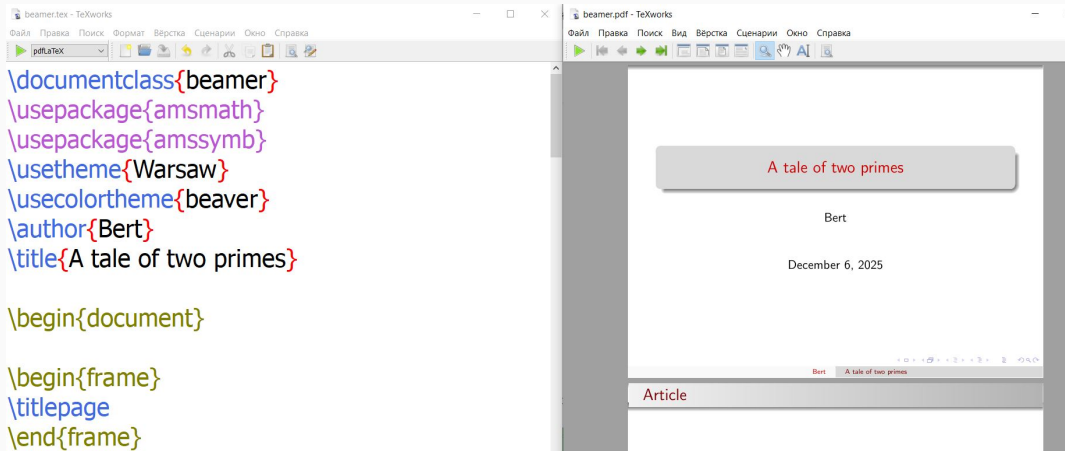


Рис. 1: pres titlepage

```
\end{block}  
\begin{block}{Euclid's theorem}  
This is a theorem.  
\end{block}  
\end{frame}
```

```
\begin{frame}{Article}  
\begin{block}{Definition}  
This is a definition.  
\end{block}  
\pause
```

```
\begin{block}{Euclid's theorem}  
This is a theorem.  
\end{block}  
\end{frame}
```

```
\begin{frame}{Title}
```



Рис. 2: pres block with pause

```
\end{block}  
\pause  
\begin{block}{Euclid's theorem}  
This is a theorem.  
\end{block}  
\end{frame}  
  
\begin{frame}{Title}  
\begin{enumerate}  
\item Element  
\pause  
\item Element  
\pause  
\item Element  
\end{enumerate}  
\end{frame}
```



Рис. 3: pres item on page


```

\begin{frame}
\frametitle{Sets}
A \alert{set} is a collection of objects.\uncover<2->{ For example:
\begin{aligned}
Z=&\{\text{cow}, \text{pig}, \text{elephant}\}.
\end{aligned}
\uncover<3->{We call the objects in $Z$ the \alert{elements} of $Z$.\uncover<4->{ We
write
\begin{aligned}
&\text{cow} \in Z
\end{aligned}
\uncover<5->{with "cow is an element of $Z$".}\uncover<6->{ Frequently encountered sets
are}
\begin{split}
&\mathbb{N} = \{1, 2, 3, \dots\} \quad (\text{"natural numbers"}) \\
&\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\} \quad (\text{"integer numbers"}) \\
&\mathbb{Q} = \{p/q : p, q \in \mathbb{Z} \text{ and } q \neq 0\} \quad (\text{"rational numbers"}) \\
&\mathbb{R} = \{\text{decimal numbers}\} \quad (\text{"real numbers"})
\end{split}
\end{frame}

```

A **set** is a collection of objects. For example:

$$Z = \{\text{cow}, \text{pig}, \text{elephant}\}.$$

We call the objects in Z the **elements** of Z . We write

$$\text{cow} \in Z$$

with "cow is an element of Z ". Frequently encountered sets are

$$\mathbb{N} = \{1, 2, 3, \dots\} \quad (\text{"natural numbers"})$$

$$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\} \quad (\text{"integer numbers"})$$

$$\mathbb{Q} = \{p/q : p, q \in \mathbb{Z} \text{ and } q \neq 0\} \quad (\text{"rational numbers"})$$

$$\mathbb{R} = \{\text{decimal numbers}\} \quad (\text{"real numbers"})$$

Рис. 4: uncover

```

\quad(\text{"integer numbers"})} \} \}
\uncover<13->{\mathbb{Q}} \uncover<14->{= \{p/q : p,q \in \mathbb{Z} \} \text{ and } q \neq 0\}} \uncover<15->{\quad(\text{"rational numbers"})} \} \}
\uncover<16->{\mathbb{R}} \uncover<17->{= \{\hbox{decimal numbers}\}} \uncover<18->{\quad\quad(\text{"real numbers"})}
\end{split}
\}
\end{frame}

\begin{frame}
  The derivative of  $f(x) = g(x) \cdot h(x)$ , with  $g(x) = x^2$  and  $h(x) = \sin(x)$  equals
  \begin{align*}
    f'(x) \uncover<2->{&= g'(x) \cdot h(x) +} \\
    \uncover<3->{&g(x) \cdot h'(x)} \} \} \uncover<4->{= 2x \cdot \sin(x) +} \uncover<5->{x^2 \cdot \cos(x).}
  \end{align*}
\end{frame}

\begin{frame}
\begin{itemize}
\item<4->  $[a,b] \uncover<5->{= \{x \in \mathbb{R} : a \leq x \leq b\}},$ 
\item<6->  $(a,b) \uncover<7->{= \{x \in \mathbb{R} : a < x < b\}},$ 
\item<8->  $(a,\infty) \uncover<9->{= \{x \in \mathbb{R} : x > a\}}.$ 
\end{itemize}
\end{frame}

```

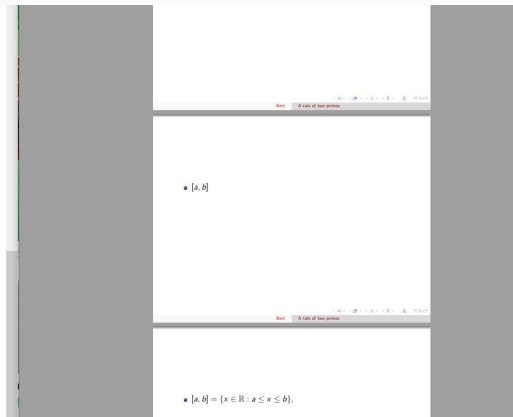


Рис. 5: item and uncover

Создание постеров. a0poster.

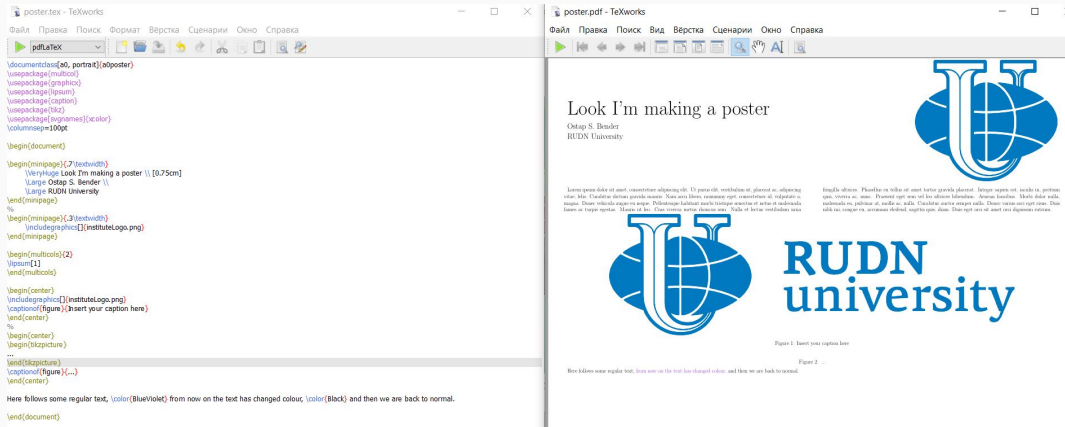


Рис. 6: poster

Создание постеров. beamerposter.

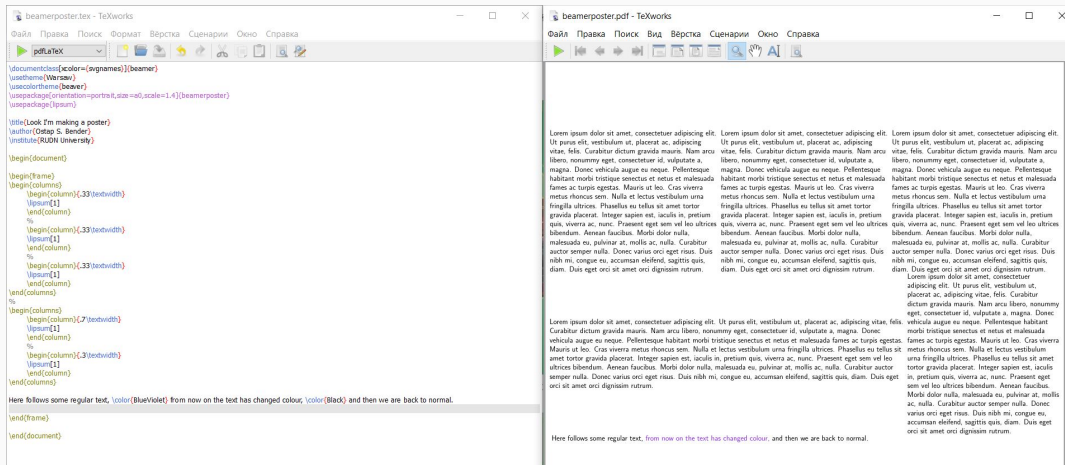


Рис. 7: beamerposter

Создание постеров. tikzposter.

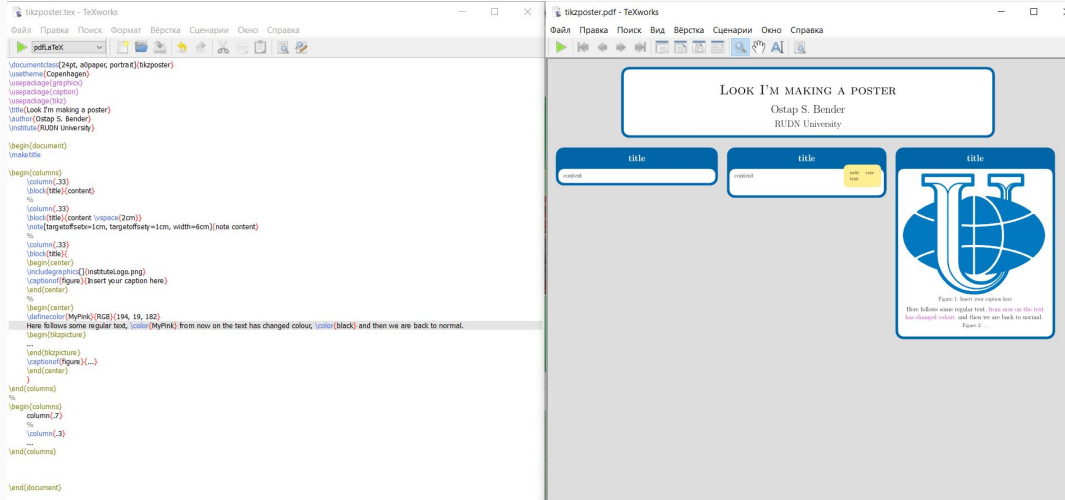


Рис. 8: tikzposter

Спасибо за внимание!